STAGE 1: Service functional/behavioral requirements and initial list of components

Functional and Behavioral requirements

The system must…

- Allow users to select teams, leagues or individual games to follow.
- Allow users to configure notification preferences by toggling alert types on or off.
- Send push notifications when relevant game events or news occur.
- Allow users to enable alerts for a single game or all games of a selected team.
- Maintain the current state of each game.
- Ensure notifications are sent only to users subscribed to that event.
- Store user preferences so they are saved over time
- Support concurrent games and concurrent users

Key Components and Responsibilities

- **Application (client) :** Allows users to create accounts, select teams and manage notification preferences. User Interface for viewing live scores and history.
- **Event Ingestion Service:** Communicates with sports data provider to receive live updates on games
- **Event Processing Service:** Processes incoming events and determines which users should receive notifications based on the users subscription and notification settings.
- **User Preference Service:** Stores user subscription data and notification settings. Maintains which teams, leagues, or games each user follows, as well as specific alert settings they have enabled.
- **Notification Service:** Sends push notifications to user device
- **Game State Management Service:** Maintains live games, including live scores, game status, and game history. This allows the system to provide up to date data for any supported game at any time.

Happy path

1) A user selects teams or games to follow and enables notifications for specific events.
2) A score change occurs during the game.
3) The system receives the event from the sports data provider.
4) The system identifies users who subscribed to that team or game
5) The system then sends notifications to the appropriate users.
6) The user receives the notification and can open the app to view more details

Failure scenarios

- **Sports data unavailable**
    - If the live sports updates are temporarily unavailable, the system should retry periodically and continue delivering notifications until a new connection is

established. During a period of disconnection, previous game information remains available.

- **Duplicate notifications**
    - If the system receives the same event multiple times, it should avoid sending duplicate notifications when possible. Occasional duplicates should not disrupt the system functionality.

- **Notification delivery failure**
    - The system should not only send notifications to users on the app or on their phones at the time a new update is received. If the phone is offline or idle, the notification should go to the user's notification history.

Figure 1 - **System Context Diagram**

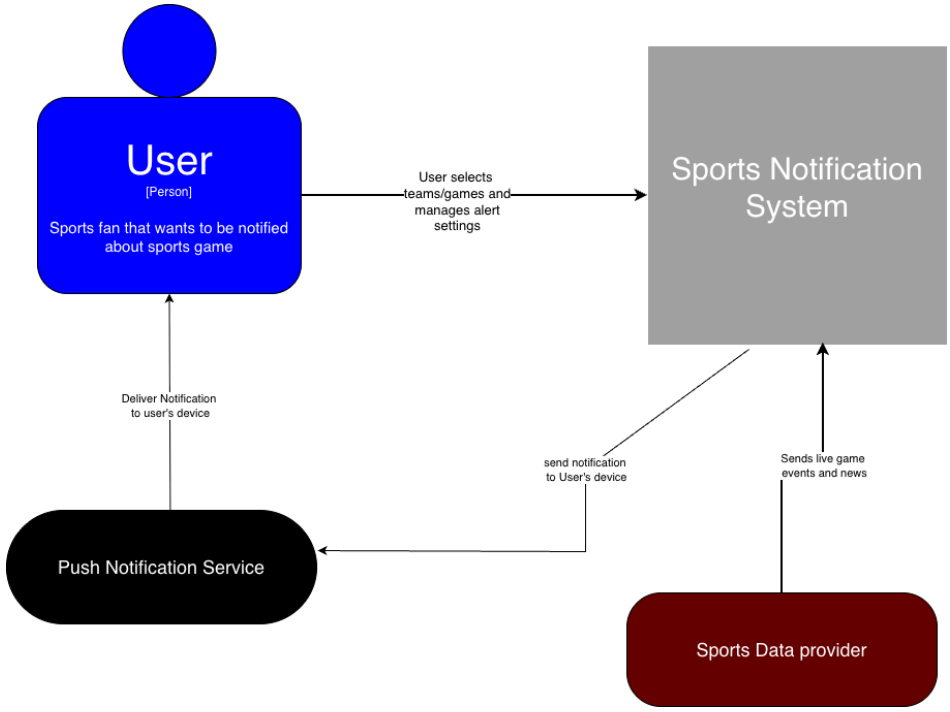


Figure 2 - **Container Diagram**

